SkillNavora

HANDBOOK 2

SQL for Salesforce Marketing Cloud

SQL fundamentals, joins, aggregates, data views and the code examples explained line by line.

Salesforce Marketing Cloud (SFMC) Academy. Verify product names, editions, licences and certifications on Salesforce's official pages.

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Edition: November 2026. Practice labs marked "account" need a provisioned Marketing Cloud account. Other labs use sample files or paper.

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Module 6: SQL for Salesforce Marketing Cloud

Level: Beginner to intermediate. SQL in SFMC: query activities, the supported syntax, aggregation, joins, data views and common audience queries.

Module 6

6.1 SQL fundamentals in SFMC

WHAT IS IT?

> SFMC SQL runs in Query Activities inside Automation Studio. It uses a T-SQL style syntax with some restrictions compared with SQL Server. Core clauses: SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY.

WHY IS IT IMPORTANT?

> Audiences, reports and suppression lists are almost always built with SQL in SFMC.

HOW DOES IT WORK AND THE STEPS

1. Write the query in the Query Activity.
2. Set a target data extension and an update option (overwrite or append).
3. Run the activity, then check the row count in the target.

REAL-WORLD EXAMPLE

> Select active customers from the Customers data extension into an Audience data extension.

COMMON ERRORS

Writing a query that works in SQL Server but is not supported in SFMC. Always test in the Query Activity.

INTERVIEW QUESTION

What is a Query Activity?

HANDS-ON EXERCISE

Write a query that selects customers in one city and counts them.

OFFICIAL DOCUMENTATION AND FURTHER READING

- SFMC SQL reference (Salesforce docs)

6.2 Aggregation, joins and CTEs

WHAT IS IT?

> Aggregates (COUNT, SUM, AVG, MIN, MAX) with GROUP BY summarise data. JOIN combines data extensions and system data views. Common table expressions (WITH) name steps. Check the official SQL restrictions for the exact features your account supports.

WHY IS IT IMPORTANT?

> Joins and aggregates produce segments and reports; a wrong join silently changes audience size.

HOW DOES IT WORK AND THE STEPS

1. Use INNER JOIN when both sides must match.
2. Use LEFT JOIN to keep all customers and show missing orders as NULL.
3. Count distinct customers, not rows, when a customer has many orders.

REAL-WORLD EXAMPLE

> Customers with no order in 90 days: LEFT JOIN orders and keep rows where the order is NULL.

CODE EXAMPLE, WITH LINE-BY-LINE NOTES

```
-- Customers with no order in the last 90 days
SELECT c.SubscriberKey, c.EmailAddress
FROM Customers c
LEFT JOIN Orders o
  ON o.SubscriberKey = c.SubscriberKey
  AND o.OrderDate ⩾ DATEADD(day, -90, GETDATE())
WHERE o.OrderID IS NULL;
-- Line by line: LEFT JOIN keeps every customer; the date condition is inside the join so only
recent orders match;
-- WHERE o.OrderID IS NULL keeps customers with no matching recent order.
```

COMMON ERRORS

Counting rows where you need distinct customers.

INTERVIEW QUESTION

Explain the difference between INNER JOIN and LEFT JOIN.

HANDS-ON EXERCISE

Write the 90-day inactivity query and explain each line.

OFFICIAL DOCUMENTATION AND FURTHER READING

- SFMC SQL reference (Salesforce docs)

Module 6

6.3 SFMC data views

WHAT IS IT?

> System data views hold sends, opens, clicks, bounces, unsubscribes, jobs, journey activity and SMS tracking. Examples: _Subscribers, _Sent, _Open, _Click, _Bounce, _Unsubscribe, _Job, _Journey, _JourneyActivity, _SMSMessageTracking, _Complaint, _BusinessUnitUnsubscribes. Each has its own keys and retention limit.

WHY IS IT IMPORTANT?

> Engagement analysis and suppression depend on these views. Retention limits mean old data may be gone, so keep your own history.

HOW DOES IT WORK AND THE STEPS

1. Check the view's documentation for keys and retention.
2. Join on SubscriberKey and JobID where the view supports them.
3. Export long-term history to your own data extension.

REAL-WORLD EXAMPLE

> Last-open date per subscriber: SELECT SubscriberKey, MAX(EventDate) FROM _Open GROUP BY SubscriberKey.

COMMON ERRORS

Assuming a data view keeps data forever.

INTERVIEW QUESTION

What is _Open used for and what limits apply?

HANDS-ON EXERCISE

List the data views you would use for an engagement report and state each key and retention limit (verify on the docs).

OFFICIAL DOCUMENTATION AND FURTHER READING

- SFMC data views (Salesforce docs)

Quick reference: SQL in SFMC

Area	Items
SQL in SFMC	SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY, JOIN types, CASE WHEN, COUNT, SUM, AVG, MIN, MAX, ROW_NUMBER. Check the SFMC SQL restrictions before use.